

AYANTIKA SARKAR

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SUMMARY

Backend Cloud Software Engineer (Java/Spring Boot/Kafka) with 4.5+ years of experience architecting highly scalable, fault-tolerant distributed systems and RESTful APIs. Proven track record of optimising high-throughput detection engines and reducing cloud operational costs. **Open to immediate relocation to the UK; actively seeking Skilled Worker Visa sponsorship roles.**

EXPERIENCE

Cloud Software Engineer, Trellix

Mar 2024 - Present

- Reduced computational overhead and AWS operational costs by 60% by architecting a high-performance detection pre-filter utilising Java & Spring Boot micro-services.
- Mentored 3 junior engineers on backend software architecture, Spring Boot best practices and advanced version control workflows with Git
- Engineered automated CI/CD pipelines leveraging Github Actions for artifact packaging and ArgoCD for declarative, GitOps-based deployments into production Kubernetes (AWS EKS) clusters
- Collaborated with teams to resolve critical customer and production issues and bugs

Associate Cloud Software Engineer, Trellix

Aug 2021 - Mar 2024

- Slashed search REST API latency by 70% by refactoring complex MySQL database queries and optimising data retrieval executional plans.
- Implemented cross-platform supportability (Linux & MacOS) within the EDR detection funnel resolving critical production bottlenecks
- Mitigated data loss in a high-throughput EDR detection funnel by 40% by implementing robust distributed event streaming logic (Apache Kafka/SpringBoot) to ensure zero-fault message delivery.
- Served as scrum master for a cross-functional Agile team of 20 engineers, streamlining sprint planning and feature delivery

Technical Intern, McAfee

Feb 2021 - Aug 2021

- Contributed to the development of new features and bug fixes for the EDR detection engine
- Re-implemented alerting mechanisms for unexpected statistical variations in the detection engine

SKILLS

Languages & Frameworks : Java(11/17/25), Spring Framework, SpringBoot, REST APIs, Micro-services

Distributed Systems & Data : Apache Kafka, MySQL, Apache Cassandra, JSON, Regex

Cloud & DevOps : AWS (EKS, EC2, S3, RDS), Docker, Kubernetes, Helm, Github Actions, ArgoCD, TeamCity

Testing & Observability : JUnit, Mockito, BDD, Apache JMeter, Grafana, Prometheus, ELK Stack

Security Engineering : Endpoint Detection & Response (EDR), YARA Tool, OCSF, OWASP Top 10, MITRE ATT&CK

AI & Developer Tools : Github Copilot, Google Gemini, Git, IntelliJ, Linux, Jira

PROJECTS

YARAWatch - Streaming Detection Engine Using YARA

Nov 2024

- Implemented a Spring Boot micro-service in Java 17 as part of a Kafka streaming pipeline to detect malicious events in real time
- Developed YARA rules to parse and identify OCSF-format JSON events, improving detection accuracy and facilitating automated threat analysis

COVID-19 Detection Using Chest X-Rays

May 2021

- Worked in a team of 4 to establish a deep-learning model for detection of COVID-19
- Achieved an accuracy of 97.62% and 96.66% for the training and testing dataset respectively

ACADEMIC PUBLICATIONS

Literature Review Of Implementation Of Machine Learning Algorithms For Improving The Network Security | [Publication](#)

Jan 2024

Serial and Parallel based Intrusion Detection System using Machine Learning | [Publication](#)

Jun 2021

Authentication and Secure Communication by Haar Cascade Classifier, Eigen Face, LBP Histogram and Variable Irreducible Polynomial in (2^8) finite field | [Publication](#)

Jun 2021

EDUCATION

Master of Technology (MTech - Part-Time), Software Systems (Specialization in Security)
Birla Institute of Technology and Science (BITS), Pilani, India

Jan 2023 - Jan 2025
CGPA:7.98

Bachelor of Technology (BTech), Information Technology
West Bengal University of Technology, Kolkata, India

Jul 2017 - Jul 2021
DGPA:8.79